

CELL: THE FUNDAMENTAL UNIT OF LIFE

→ Biology is the study of living.

- Zoology - Study of animals
- Botany - Study of Plants

→ Discovery of cell:-

- Robert Hooke in the year 1665, made a chance observation through self designed microscope of a substance that he took from the Oak tree called the cork. Cork is a substance that comes from the bark of the tree.
- Hooke observed these small boxes like structures like little rooms. These little rooms were called 'cell'. In Latin 'cell' means 'little room'.
- Leeuwenhoek (1647) with the help of an improved microscope discovered 'living cell' for the first time in pond water.
- Robert Brown in 1831 discovered the nucleus of the cell.
- Purkinje ~~discovered~~ named the fluid substance of the cell 'Protoplasm' in 1839.
- All living organisms are made up of cells, (unicellular) or (multicellular). Cells are the smallest unit of life of life that can carry out all the functions necessary for life.

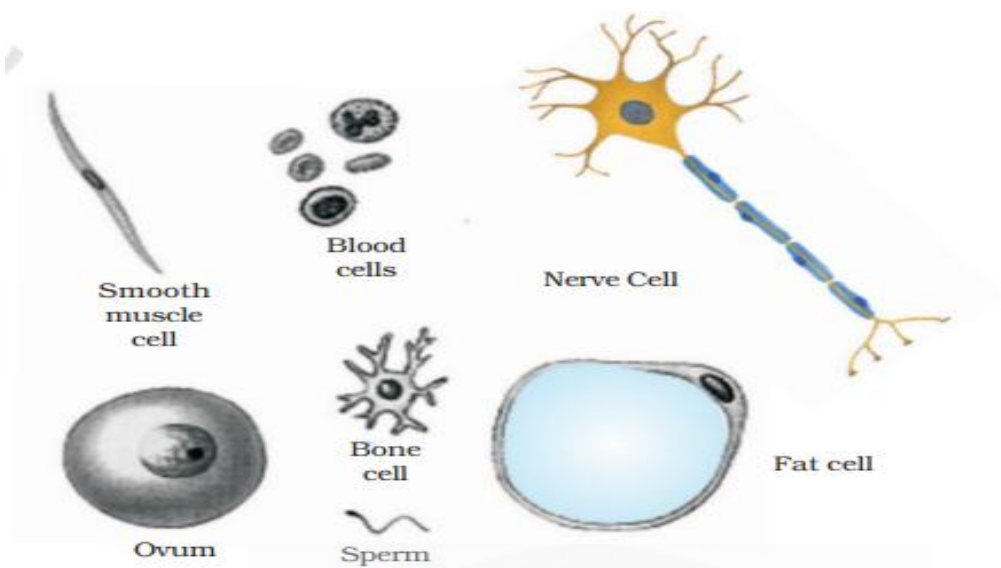


Fig. 5.3: Various cells from the human body

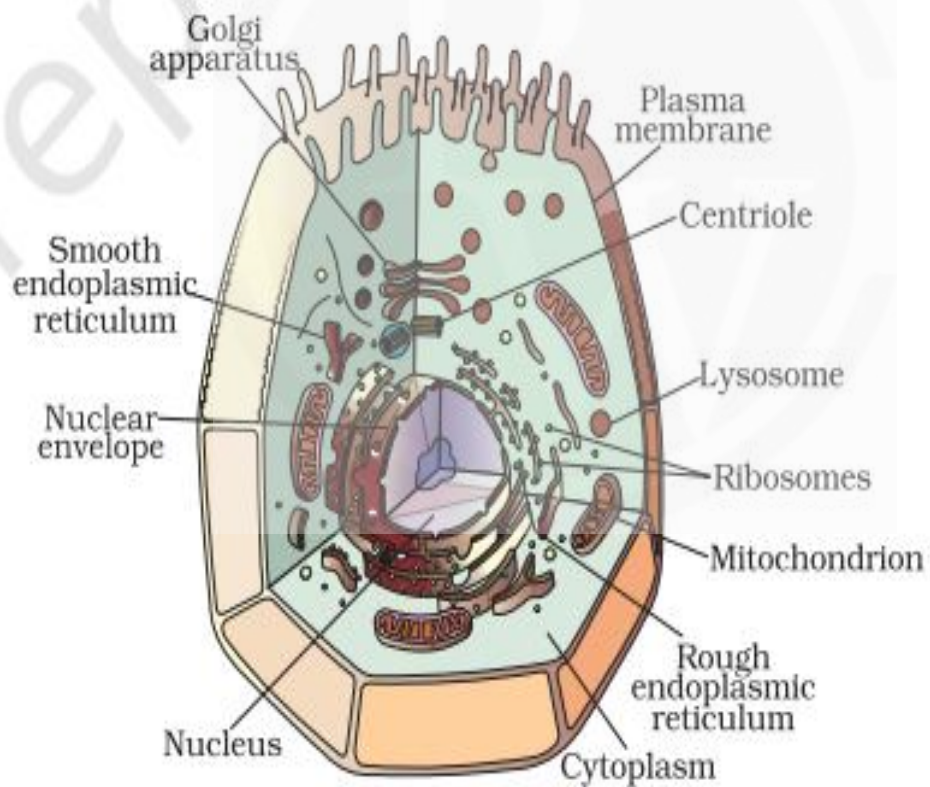


Fig. 5.5: Animal cell

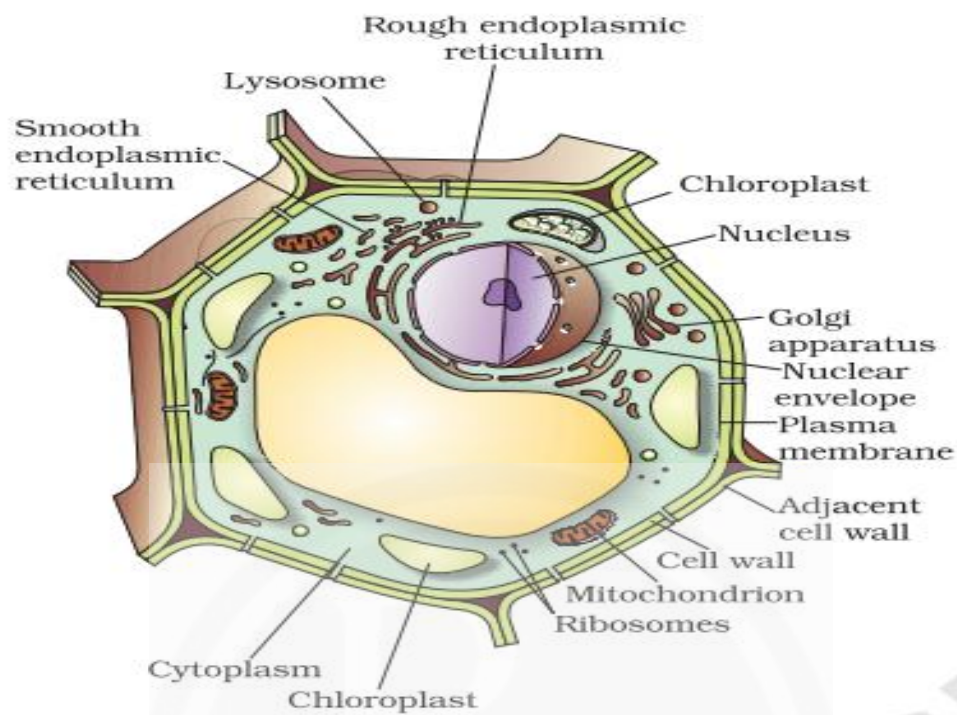


Fig. 5.6: Plant cell

Cell Theory:-

- Cell theory is the fundamental concept in biology that describes the properties and functions of cells, the basic unit of life. It was developed in the 19th century by scientists Matthias Schleiden (Botanist) & Theodor Schwann (Zoologist).
- Cell Theory is as - All living organisms are made up of cells.

Cell Lineage Theory:-

- All cells arise from pre-existing cells (Omnis - cellula e cellula):- This principle, proposed by Rudolf Virchow, states that new cells are formed by division of existing cells. Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.

Cell Theory is understood as:-

- (i) All living organisms are composed of cells and products of cells.
- (ii) All new cells arise from pre existing cells.

- Cell Size - Mycoplasma, the smallest cells, are only $0.3\mu\text{m}$ in length.
- The largest isolated single cell is the egg of an ostrich.
- Among multicellular organism, human red blood cells are about $7.0\mu\text{m}$ in diameter.
- nerve cells are some of the longest cells

→ Cell diversity and division of labor -

- Cell diversity is the variety of cell types that make up tissues and organisms. The human body has about 200 types of cells. They make for various functions in the body.

The cells make be disc shaped, polygonal, columnar, cuboid, thread like or even irregular. The shape may vary with the function they perform.

→ In biology, division of labour - refers to the distribution of different functions among different parts or different cells in an organism. DOL can be seen generally in multicellular organisms.

Cell Structure :-

If we look under the microscope, we would come across three features in almost all cells - plasma membrane, nucleus and cytoplasm.

→ Plasma membrane or cell membrane :-

→ surrounds the cytoplasm of all cells, separating the internal contents of the cells from its external environment

→ mainly composed of lipids and proteins and cytoplasm.

→ Fluid Mosaic Model - by given by Singer and Nicolson (1972). According to this, the quasi fluid nature of lipids enables lateral movement of proteins within the overall bilayer.

→ The major lipids are phospholipids that are arranged in a bilayer.

→ The lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tail towards (non polar tails of saturated hydrocarbon).

→ Membrane proteins can be classified as integral proteins and peripheral proteins. Peripheral proteins lie on the surface of membrane while integral proteins are partially or totally buried in the membrane.

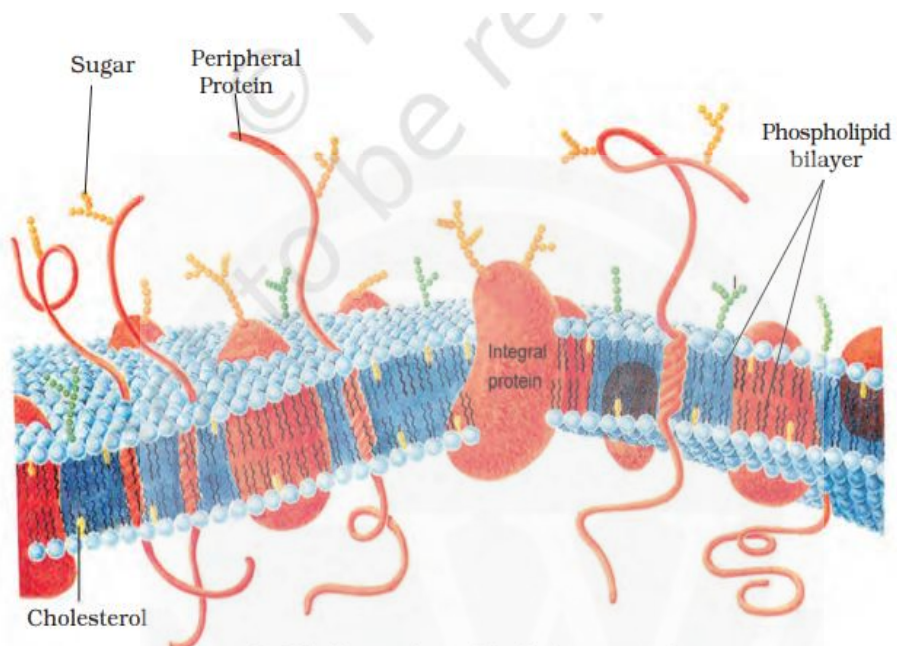


Figure 8.4 Fluid mosaic model of plasma membrane

The fluid nature of the membrane is also important from the point of view of functions like cell growth, formation of intercellular junctions, secretions, endocytosis, cell division etc.

→ Functions of Cell Membrane or plasma membrane:-

① Selective Permeability - The membrane controls the movement of substance into and out of the cell.

② Transport - The membrane contains proteins that facilitate the transport of ions, nutrients and waste products. This includes:-

a) Passive transport - that does not require energy. Neutral solutes may move across the membrane by the process of simple diffusion along the concentration gradient. Movement of water by diffusion is called ~~diffusion~~ osmosis.

b) Active Transport - As the polar molecules cannot pass through the non polar lipid bilayer they require a carrier protein of the membrane to facilitate their transport across the membrane. Such a transport is energy dependent process in which ATP is utilised and is called active transport.

→ Cell Wall:-

→ The cell wall is rigid, non living layer that surrounds the plasma membrane of a cell.

→ Composition - (i) Plants:- (made up of cellulose, hemicellulose, pectins and proteins).

(ii) Algae - (made up of cellulose, galactans, mannans and minerals like calcium carbonate).

(iii) Fungi - (made up of Chitin).

The cell wall of a young plant cell, the primary wall is capable of growth which gradually diminishes as the cell matures and secondary wall is formed on the inner (towards membrane) side of the cell.

Functions of Cell wall:-

- ① Protection - It protects the cell ~~wall~~ from physical damage infection and osmotic bursting
- ② Structure - It gives the cell shape and structure
- ③ Regulation - It regulates the flow of information between the cell and the expression
- ④ Transport - It allows elements to pass through the cell from a permeable membrane.

Middle Lamella - The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together. The cell wall and middle lamella may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cell.

- Endomembrane System:- Each member of the endomembrane system has distinct function and structure.
- It includes → endoplasmic reticulum, golgi complex, lysosomes, vacuoles.
 - Since chloroplast and mitochondria are not coordinated with the upper green organelles they are not included in the endomembranous system.

① Endoplasmic Reticulum :-

- This is a complex network of tubes called cisternae, the lumen of these cisternae is filled with fluid called endoplasmic reticulum.
- ER has three parts → Cisternae, Tubules & Vesicles.
- Cisternae are long inter-connected, flattened sac like structures lying parallel to each other in the form of pile or ~~stack~~ stack.
- Tubules are minute branches like structure attached to cisternae.
- Vesicles are bubble like structures produced at the ends of tubules and are finally pinched off to get attached with Golgi complex.

⇒ Types of ER → ① Smooth Endoplasmic Reticulum:- also called as agranular endoplasmic reticulum — without ribosomes. This ER is meant to synthesize carbohydrates and lipids.

② Rough Endoplasmic Reticulum → bears ribosomes to its surface, so it is also called as granular endoplasmic reticulum. This ER is meant to synthesize proteins.

- The functions of the ER are to provide and form the skeletal framework of the cell, to provide a pathway for the distribution of nuclear material from one cell to the other and to synthesize fats, steroids and cholesterol with the help of enzymes secreted by the cell.



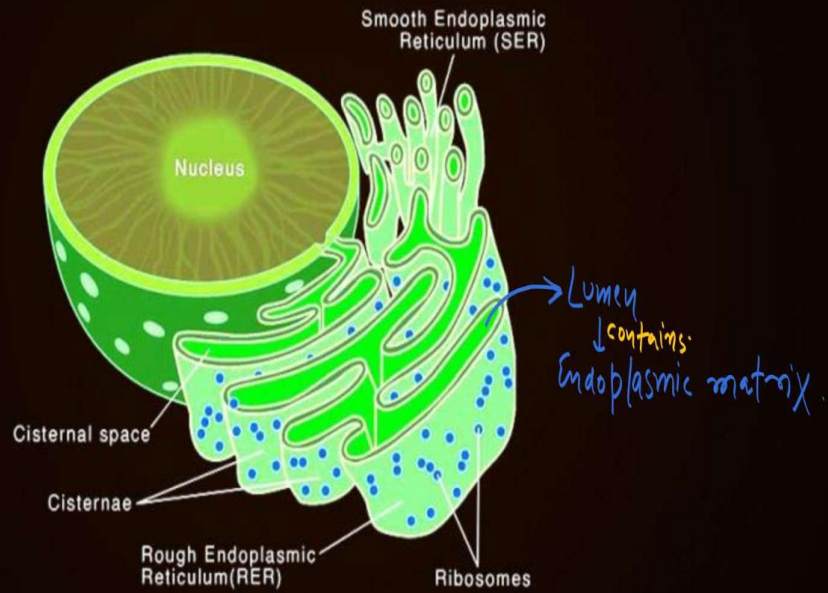
Endoplasmic Reticulum

→ Discovery :- Porter.



Parts:-

- ① Cisternae
- ② Tubules.
- ③ Vesicles.



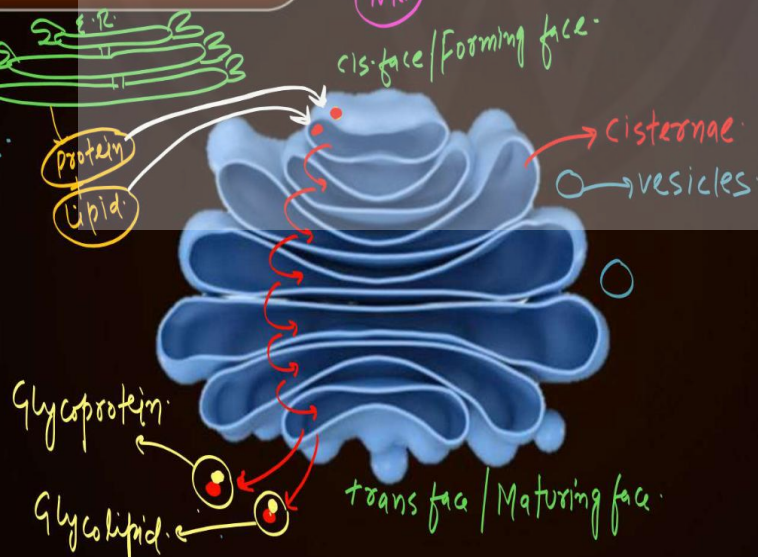
Golgi Apparatus

Discovery:- Camilo Golgi.



Parts:-

- ① Cisternae.
- ② Vesicle.



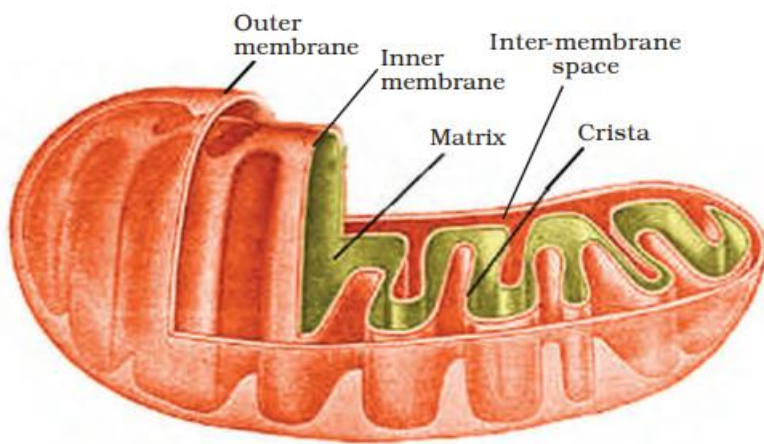


Figure 8.7 Structure of mitochondrion (Longitudinal section)

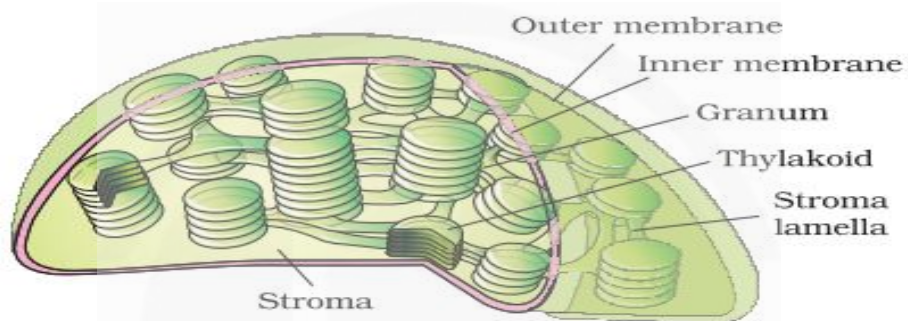


Figure 8.8 Sectional view of chloroplast

- Plastid - discovered by Ernst Haeckel
- named by Ernst Haeckel
- present in plant cell and animal cell - it is absent

Types:-

- ① Chloroplast - green plastid
- contain chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis.
- ② Chromoplast - fat soluble carotenoid pigment like carotene, xanthophyll.
- ③ Leucoplast - The leucoplast are the colorless plastids of varied shapes and sizes that store nutrients

- Leucoplast
- ① Amyloplast - store carbohydrates (starch) e.g. potato.
 - ② Elaioplast - store oil and fats
 - ③ Aleuroplast - store proteins.

- Like mitochondria, the chloroplasts are also double membrane bound.
- The space limited to the inner membrane of chloroplast is called stroma
- A number of organised flattened membranous sacs called thylakoids are present in the stroma.
- The thylakoids are arranged in stacks like the piles of coins called grana. They are also flat membranous tubules called the stroma lamellae connecting the thylakoid of different grana.
- The membrane of thylakoid enclose a space called lumen in which chlorophyll is present.
- It also has small, double stranded circular DNA molecules and ribosomes.

Golgi Apparatus → first observed densely stained reticular structures near the nucleus. These were later named Golgi ~~app~~ bodies after him.

→ They consist of many flat, disc shaped sacs or cisternae and are stacked parallel to each other. Varied number of cisternae are present in a Golgi Apparatus or complex.

→ The Golgi cisternae are concentrically arranged near the nucleus. with distinct convex cis or the forming and concave trans or the maturing face.

→ Golgi apparatus is the important site of formation of glycoprotein and glycolipids.

→ A number of proteins synthesised by ribosomes in the endoplasmic reticulum are modified in the cisternae of the golgi apparatus before they are released from its trans face.

Lysosomes :- the membrane bound vesicular structure formed by the process of pinching in the golgi apparatus.

→ The isolated lysosomal vesicle have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases like lipases, proteases, carbohydrases), optimally active at the acidic pH.

→ These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

Mitochondria was first discovered by Albert von Kolliker.
The name mitochondria was given by Carl Benda.

- Powerhouse of cell
- Semi-autonomous cell organelle

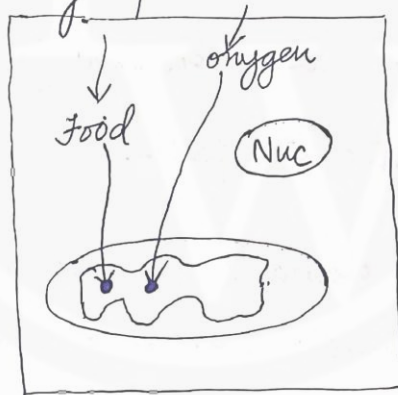
→ 1st observed by Kolliker in 1880 but Benda in 1897 gave the present name of mitochondria

→ Shape, number and Size - cylindrical or sausage shaped and distributed in variable number depending upon the physiological activity of the cells and having a diameter of $0.5\text{ }\mu\text{m}$ and length $2\text{ }\mu\text{m}$ (average).

→ Each mitochondria is bound by a double membrane, the outer membrane and the inner membrane dividing its lumen into two aqueous compartments

→ The inner membrane is folded into ridges called cristae, which increase the surface area of the membrane. The inner membrane on its inner side possesses small ~~in~~ tennis racket particles or lollipop like structure called elementary particles or FOF1 particles or oncosomes

ATP - Adenosine
Tri-phosphate



The space inside the mitochondria → is called as inner compartment is filled with a homogeneous substance called matrix which possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins.

→ The two membranes have their own specific enzymes associated with the mitochondria function. Mitochondria has sites of aerobic respiration. They produce cellular energy in the form of ATP, hence are called power house of the cell.

Ribosomes :- Smallest cell organelle.
discovered by George Emil Palade (1953)
membrane less cell organelle.

- Made of protein & RNA (Ribonucleic acid) (N)
8888888
polysome
- Has two subunits → larger subunit and smaller subunit
- Has two type { 80S ribosome and 70S ribosome
{ S = Svedberg unit = sedimentation coefficient.
- Also known as protein factory of a cell because it helps in protein synthesis.
- 80S → eukaryotic ribosomes and 70S are prokaryotic ribosomes.
- Each ribosome has two units — larger and smaller subunit.
- 80S → 60S and 40S subunits.
- 70S → 30S and 50S subunits
- Sedimentation coefficient is indirectly a measure of density and size.

Vacuole :- Present in plant cells and absent in animal cell, if present will be of small size.

- In plants cell vacuole occupies 90% space.
- Vacuole is bounded by a single membrane called tonoplast.
- In plants the tonoplast facilitates the transport of a number of ions and other materials against the concentration gradient into the vacuole.
- In Amoeba the contractile vacuole is important for osmoregulation and excretion. In many cells, as in protista, food vacuoles are formed by engulfing the food particles.

→ Centrosome and Centrioles :-

- Centrosome is an organelle usually containing two cylindrical structures called centrioles.
- They are surrounded by amorphous pericentriolar materials. Both the centrioles in a centrosome lie perpendicular.
- Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like a cartwheel.
- Each of the peripheral fibril is a triplet. The adjacent triplets are ~~the~~ also linked (AC linker).
- The central part of the proximal region of the centriole is also proteinaceous and called the hub, which is connected with tubules of the peripheral triplets by radial spokes made of protein. The centrioles form the basal body of cilia or flagella and spindle fibers that give rise to spindles apparatus during cell division in animal cells.

→ Microbodies :-

- They are small single membrane bound minute vesicles or cell organelles that contain various enzymes and are present in both plant and animal cell. Examples are Peroxisomes and Glyoxysomes.
- Function :- Their primary function include the breakdown of fats, alcohol and amino acid. In animal cell, they metabolism in number of toxic substance.

→ Cytoskeleton :-

- An elaborate network of filamentous proteinaceous structures consisting microtubules, microfilaments and intermediate filaments present in the cytoplasm.
- Functions - Such as mechanical support, mobility, maintenance

of the shape of the cell.

→ **Nucleus**: [Robert Brown (discovery)
Boss of a cell.

→ It is highly extended thread like structure called chromatin, nuclear matrix and one or more spherical bodies called nucleoli.

→ Chromatin was stained by basic dyes by Flemming.

→ Electron microscopy has revealed that the nuclear envelope has two parallel membranes with a space between called the per-nuclear space.

→ The outer membrane usually remains constant continuous with the endoplasmic reticulum and also bears ribosomes on it.

→ At a number of places the nuclear membrane or envelope is interrupted by minute pores which are formed by the fusion of its membranes called nuclear pores. which are the passages through which movement of RNA and protein molecules take place in both directions between the nucleus and the cytoplasm.

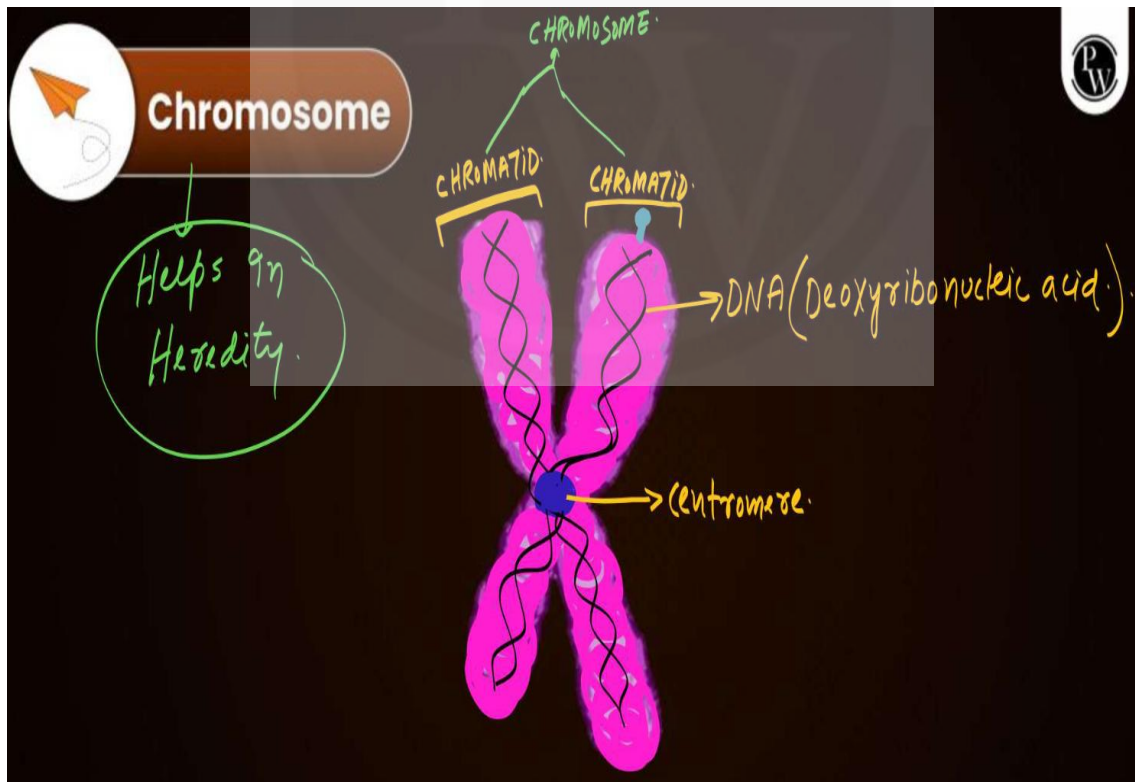
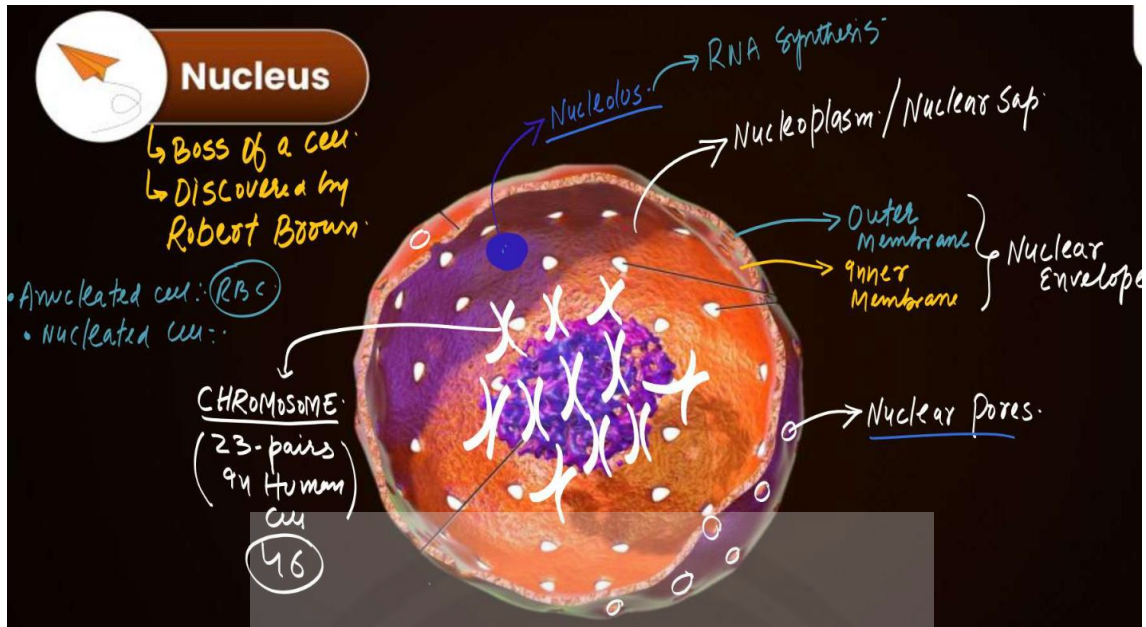
→ Mature (RBCs) erythrocytes of many mammals and sieve tube cells of vascular plants lack nucleus. The nuclear matrix or the nucleoplasm contains nucleolus and chromatin.

→ The nucleoli are spherical structures present in the nucleoplasm and is not a membrane bound structure. It is the site of active ribosomal RNA synthesis. Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

→ Nucleus has loose and indistinct network of nucleoprotein fibres called chromatin.

Chromatin contains DNA and some basic proteins called histones, some non basic ^{histone} proteins and also some RNA.

→ Chromosome helps in heredity.



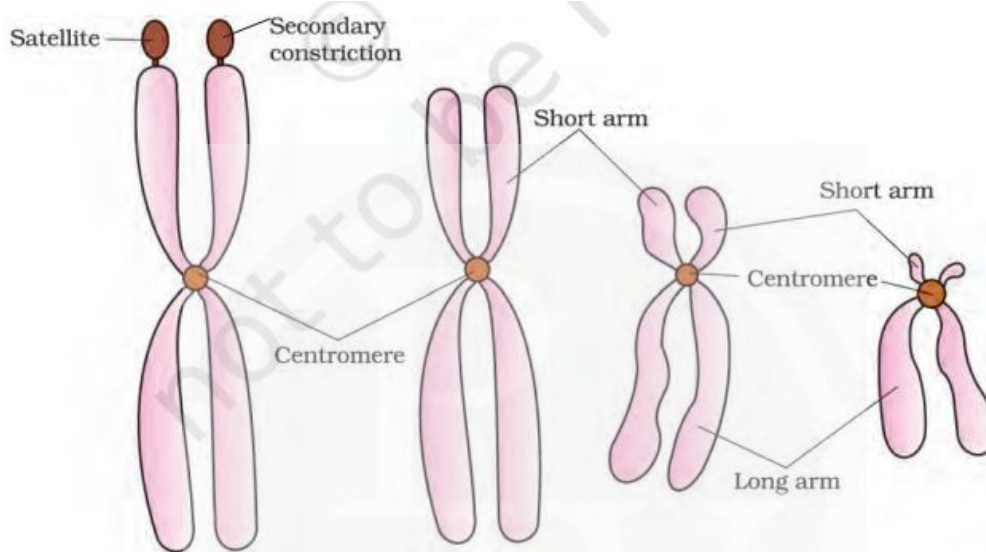


Figure 8.13 Types of chromosomes based on the position of centromere

→ Every chromosome has a primary constriction or the centromere on the sides of which disc shaped structures called kinetochores are present.

→ Centromere holds two chromatids of a chromosome. Based on the position of the centromere, the chromosome can be classified into four stages —

- ① Metacentric chromosome - has middle centromere forming two equal arms of the chromosome.
- ② Sub-metacentric chromosome - has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm.
- ③ Acrocentric chromosome - the centromere is situated close to its end forming one extremely short and one very long arm.
- ④ Telocentric chromosome - has a terminal centromere.

→ Prokaryotic Cells:-

- represented by bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia like Organism).
- Smaller than Eukaryotic cells. They have varied shapes and sizes.
- The four basic shapes of bacteria are bacillus (rod like), coccus (spherical), vibrio (comma), spirillum (spiral).
- All prokaryotic cells have a cell wall surrounding the membrane except in mycoplasma.
- The semi fluid matrix filling the cell is the cytoplasm. There is no well-defined nucleus.
- The genetic material is naked, not enveloped by a nuclear membrane.
- In addition to genomic DNA, there are small circular DNA outside called plasmids. The plasmid DNA confers certain unique phenotype characters to such bacteria. One such character is resistance to antibiotic.
- no organelles except ribosomes are found. Prokaryotes have something unique in the form of inclusions.
- A differentiated form of cell membrane called mesosome is the characteristic of prokaryotes.

→ Cell Envelope and its Modifications:-

- The cell envelope consists of tightly bounded three layered structure:-

- ① The glycocalyx - it is the outermost structure which has different composition and thickness among different bacteria. It could be loose sheath called slime layer in some, while in others it could be thick and tough, called the capsule.
- ② The cell wall determines the shape and provides strong structural support to prevent the bacterium from bursting or collapsing.
- ③ Plasma membrane is selectively permeable in nature and interacts with the outside world.

→ A special membrane structure is the mesosome which is formed by the extensions of plasma membrane. These extensions are in the form of vesicles, tubules and lamellae— These help in the formation of cell wall, DNA replication, distribution of daughter cells. Also help in respiration, secretion and enzymatic content.

→ In some cyanobacteria there are other membranous extensions called the chromatophores which contain pigments.

→ Bacteria may be motile or non motile. If motile, it has a thin filamentous extensions from their cell wall flagella.

→ Bacterial flagellum has three parts— filament, hook and basal body.
→ The filament is the longest portion and extends from cell surface to the outside.

→ Pili and Fimbriae are also surface structures but do not play a part in the mobility. The pili are elongated tubular structures made of special protein. The fimbriae are small bristle like fibers. In some bacteria they are known to help attach the bacteria to the rock in streams and also to the host tissues.

→ Ribosome and inclusion bodies :-

→ Ribosomes are associated with the plasma membrane of the cell in prokaryotes. They are about 15nm by 20nm in size and are made up of two sub units 50S and 30S. which are present together as 70S.

→ There are sites of protein synthesis and may attach to a single mRNA and form a chain called polyribosomes or polysome.

→ The ribosome of a polysome translate

Inclusion bodies - Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm - i.e. phosphate granules, cyanophycin granules and glycogen granules.

→ On the basis of difference in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram's -

- (i) Gram positive - those that take up the stain.
- (ii) Gram negative - those that do not take up the stain.

Prokaryotic Cells

→ Prokaryotic cells are simple unicellular organism that lack a defined nucleus and membrane bound organelles.

→ e.g. Bacteria, Archaea.

→ Nucleus - no true nucleus, the genetic material (DNA) is found in the region called the nucleoid.

→ Do not have membrane-bound organelles like mitochondria.

→ Ribosome - present, but smaller (70S)

→ Cell wall - present in most, composed of peptidoglycan (in bacteria)

→ Reproduction - Asexual reproduction by binary fission

Eukaryotic Cells

→ Eukaryotic cells are more complex and found in multicellular organism as well as some unicellular organism (e.g. protists).

e.g. Plants, animals, fungi, protists.

→ Nucleus - have a well defined, membrane bound nucleus that contain the cell's genetic material.

→ Contain membrane bound organelles such as mitochondria, endoplasmic reticulum, Golgi apparatus and chloroplast.

→ Ribosome (Larger) - 80S

→ Cell wall - Present in plants (cellulose) and chitin, but absent in animal cells.

→ Reproduction - Asexual (mitosis) and sexual (meiosis).

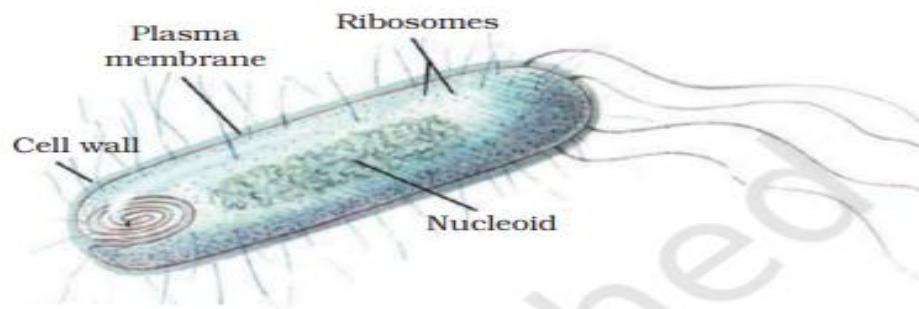


Fig. 5.4: Prokaryotic cell

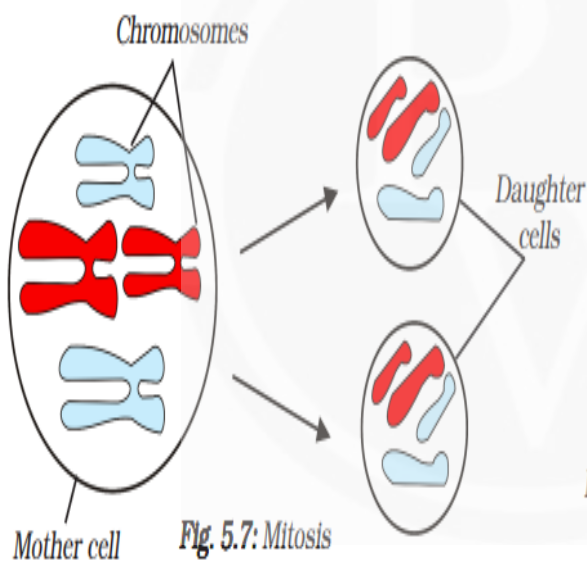


Fig. 5.7: Mitosis

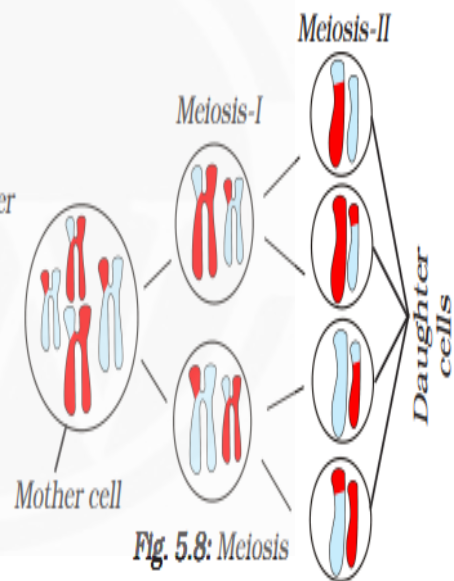


Fig. 5.8: Meiosis

→ Cell Division :- new cells are formed in organisms to grow, to replace old, dead and injured cells, and to form gametes required for reproduction.

→ The process by which ^{new} cells are formed is called cell division.
→ There are two types of cell division → mitosis and meiosis.

→ Mitosis → cell divide for growth
→ each cell called mother cell divides to form two identical daughter cells.

→ The daughter cells have the same number of chromosome as mother cell. It helps in growth and repair of tissues.

→ Meiosis → The ~~rep~~ specific cells called reproductive cells or tissues in animals and plants divide to form gametes, which after fertilization give rise to offspring.

→ They divide by a different process called meiosis which involves two consecutive division.

→ When cell divides by meiosis it produces four new cells only.

→ The new cell only have half the number of chromosome than that of mother cells.